

ANNEXURE – I

COVER SHEET (to be completed and attached as the first page of the submission)

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Programme / Branch	B.tech CSE AI&ML
Semester / Section / Batch	Sem-V, Section-C
Course Code and Title	AIM3102 / Automata and Formal Languages
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Course Coordinator	Dr. Shinnu Jangra, Department of AI & ML, MUJ
Marks Awarded (out of 20)	Scaled Marks (out of 10):
Signature of the Evaluator	

Declaration by the Student

I declare that the work submitted in this assignment is my own. I have not copied it, in part or in full, from any classmate, book, website or generative-AI tool without acknowledgement, and I understand that any such copying will attract zero marks under the University's academic integrity policy.

Date: 28/09/2026

Signature of the Student: 

a) $\Sigma = \{0, 1\}$

i) $\Sigma^0 = \{\epsilon\}$

ii) $\Sigma^1 = \{0, 1\}$

iii) $\Sigma^2 = \{00, 01, 10, 11\}$

iv) $|\Sigma^n| = |\Sigma|^n$

$$|\Sigma^5| = 2^5 = 32$$

- Kleene star, $\Sigma^* = \{\text{all finite strings over } \Sigma\}$
it includes ϵ .

for eg: $\{0, 1\}^* = \{\epsilon, 0, 1, 00, 01, 10, 11, \dots\}$

- Positive closure, $\Sigma^+ = \{\text{all non empty finite string over } \Sigma\}$
it not includes ϵ

for eg: $\{0, 1\}^+ = \{0, 1, 00, 10, \dots\}$

$$\Sigma^+ = \Sigma^* - \{\epsilon\}$$

- Length of a string: for eg: $w = 01011$

$$|w| = 5 \text{ \& } |\epsilon| = 0$$

- Concatenation,

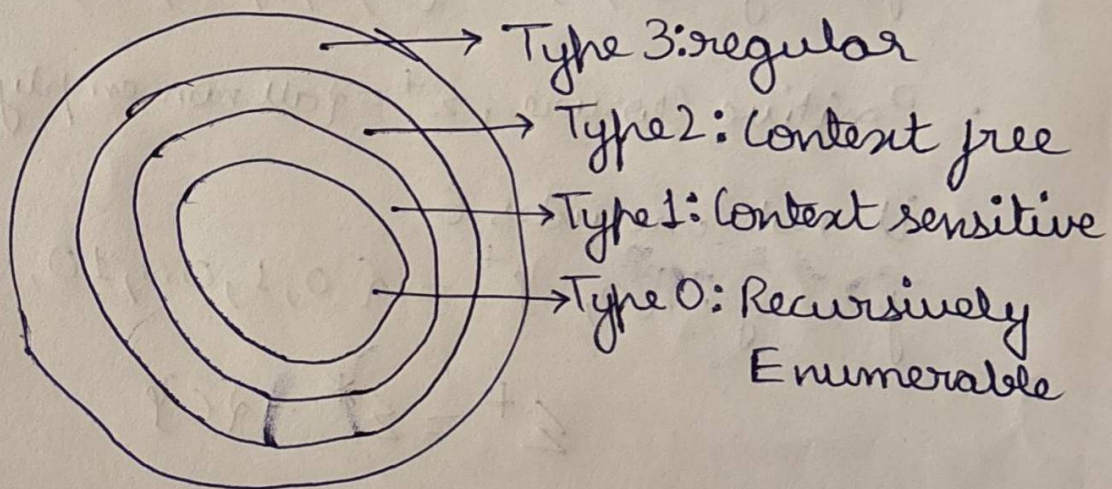
$$x = 01 \text{ \& } y = 101$$

$$xy = 01101$$

b) Chomsky Hierarchy

$$\text{Regular} \subsetneq \text{CFL} \subsetneq \text{CSL} \subsetneq \text{RE}$$

<u>Type</u>	<u>Grammar</u>	<u>Recognizing Machine</u>
0	Unrestricted	Turing Machine
1	Context Sensitive	Linear bounded automaton
2	Context free	Pushdown Automata
3	Regular	Finite Automata



Q2) DFA ending in 01

$$\Sigma = \{0, 1\}$$

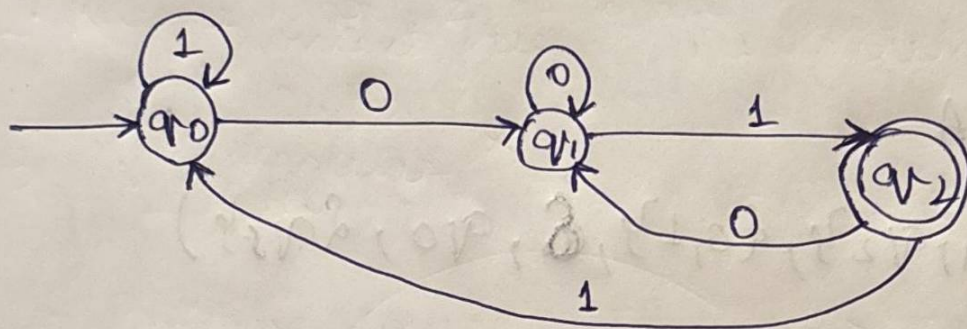
$$M = \{Q, \Sigma, \delta, q_0, F\}$$

$$Q = \{q_0, q_1, q_2\}$$

$$F = \{q_2\}$$

Transition Function :-

State	0	1
q_0	q_1	q_0
q_1	q_1	q_2
q_2	q_1	q_0



Trace for 11001

- $q_0 \xrightarrow{1} q_0$
- $q_0 \xrightarrow{1} q_0$
- $q_0 \xrightarrow{0} q_1$
- $q_1 \xrightarrow{0} q_1$
- $q_1 \xrightarrow{1} q_2$

b) Number is divisible by 3:

- (1) q_0 : number of 1s = $0 \bmod 3$
- (2) q_1 : number of 1s = $1 \bmod 3$
- (3) q_2 : number of 1s = $2 \bmod 3$

q_0 is final state

Transition Table:-

State	0	1
q_0	q_0	q_1
q_1	q_1	q_2
q_2	q_2	q_0

Q3) NFA to DFA

$$n = (\{q_0, q_1, q_2\}, \{0, 1\}, \delta, q_0, \{q_0\})$$

$$\delta(q_0, 0) = \{q_0, q_1\}$$

$$\delta(q_0, 1) = \{q_0\}$$

$$\delta(q_1, 1) = \{q_2\}$$

$$A = \{q_0\}$$

DFA State	NFA Subset	0	1
A	$\{q_0\}$	B	A
B	$\{q_0, q_1\}$	B	C
C*	$\{q_0, q_2\}$	B	A

• the NFA reaches q_2 when,

$$0 \rightarrow q_1 \rightarrow 1 \rightarrow q_2$$

DFA states, A, B, C, only C is final
thus, (A,B), (A,C), (B,C) all are distinguishable

minimum DFA has 3 states

for n states NFA, the subset = 2^n

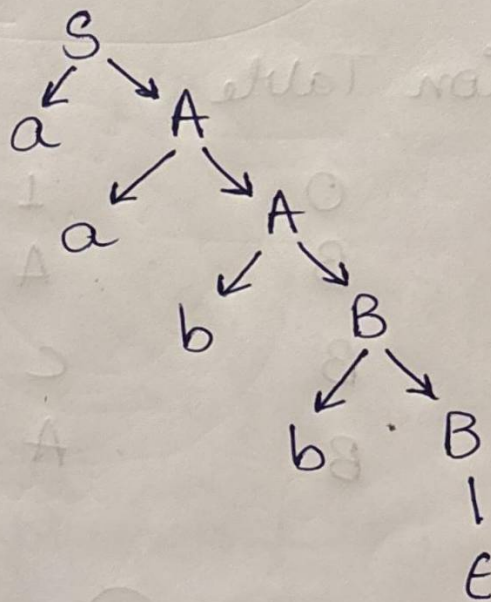
Q4) $L = \{w \in \{a,b\}^* : w \text{ contains substring } ab\}$

$$S \rightarrow aA | bS$$

$$A \rightarrow aA | bB$$

$$B \rightarrow aB | bB | \epsilon$$

for a a b b



b) DFA to right linear grammar

- $q_0 \rightarrow S$
- $q_1 \rightarrow A$
- $q_2 \rightarrow B,$

transition give,

$$S \rightarrow 0A \mid 1S$$

$$A \rightarrow 0A \mid 1B$$

$$B \rightarrow 0A \mid 1S \mid \epsilon$$

Q5) i) a) Beginning with 1 and ending with 0

$$1(0+1)^*0$$

$$b) (0+1)^*00(0+1)^*$$

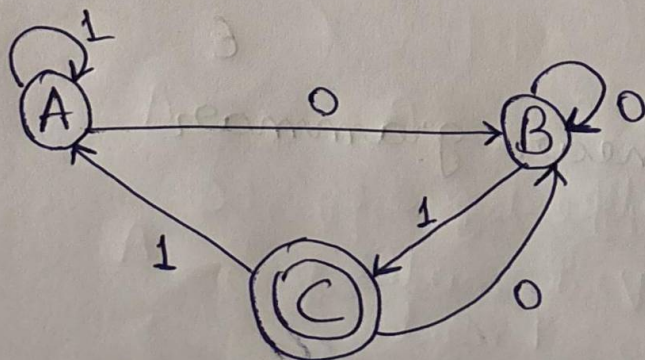
$$c) ((0+1)(0+1))^*$$

ii) DFA

Transition Table

State

	0	1
A	B	A
B	B	C
C	B	A



$$a) (r+s)^* = r^* s^*$$

this is false,

$$\text{take, } r=a, s=b$$

$$\text{then, } ba \in (a+b)^*$$

$$\text{but, } ba \notin a^* b^*$$

$$\therefore, (r+s)^* \neq r^* s^*$$

$$b) (r^*)^* = r^*$$

this is true.

$$Q6) L' = \{w \in \{a,b\}^* : n_a(w) = n_b(w)\}$$

Consider, Regular Language,

$$R = a^* b^*$$

$$\therefore, L' \cap R$$

$$\text{But, } L' \cap R = \{a^n b^n : n \geq 0\}$$

$\{a^n b^n : n \geq 0\}$ is not regular

$\therefore$, our assumption is false,

Hence, L' is not regular

for a DFA, $M = (Q, \Sigma, \delta, q_0, F)$

i) Emptiness, $L(M) = \phi$

ii) Finiteness, $L(M)$ is infinite

- if a reachable cycle exists that can reach $\frac{1}{2}$ some final states.

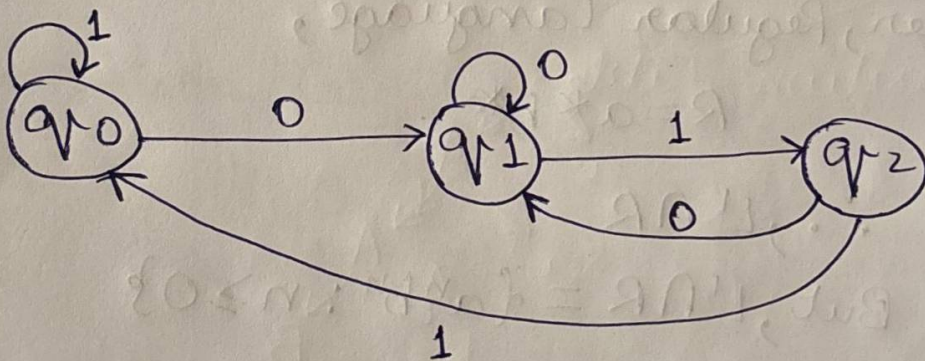
otherwise, $L(M)$ is finite

Q7) a) $L = \{w : |w| \bmod 3 = 0\}$

Transition Table

State

	0	1
q_0	q_1	q_1
q_1	q_2	q_2
q_2	q_0	q_0



b) $L = \{w : n_a(w) \bmod 3 > n_b(w) \bmod 3\}$

we need q 's states,

q_{ij} where,

$$i = n_a(w) \bmod 3$$

$$j = n_b(w) \bmod 3$$

out ; q_{100}

on a , $a_{ij} \xrightarrow{a} q_{(i+1) \bmod 3, j}$

on b , $a_{ij} \xrightarrow{b} w_{i, (j+1) \bmod 3}$

$\therefore, i > j$ therefore, $F = \{q_{10}, q_{20}, q_{21}\}$

c)

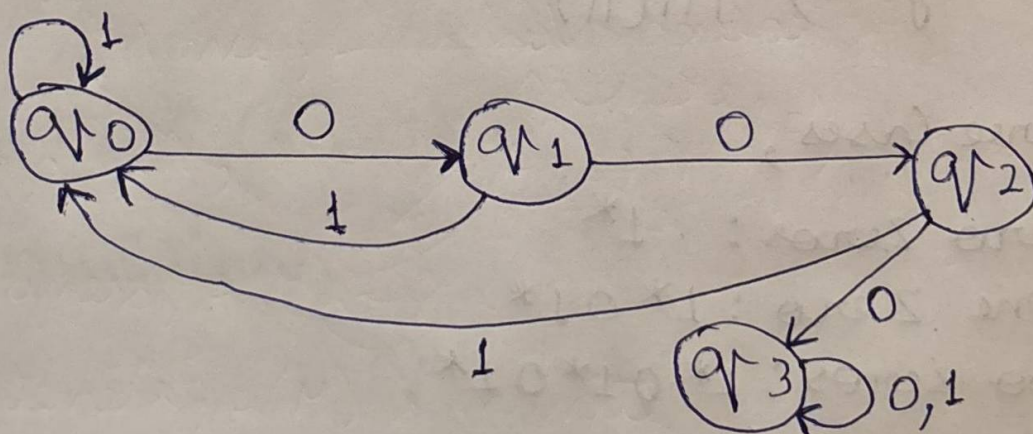
c) False

c) False

Q8) Every 00 is followed by 1

Transition Table:

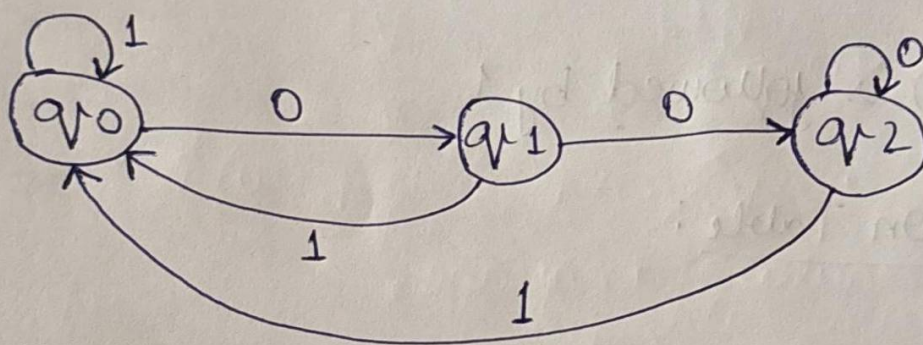
State	0	1
q_0	q_1	q_0
q_1	q_2	q_0
q_2	q_3	q_0
q_3	q_3	q_3



Q9) DFA for $(0+1)^*00$

Transition Table,

State	0	1
q_0	q_1	q_0
q_1	q_2	q_0
q_2	q_2	q_0



Q10) a)

Set $\{1^{2n+1} : n > 0\}$

since $n > 0$, possible lengths are:

3, 5, 7, 9, ...

therefore, $111(11)^*$

b) Three cases,

Zero Zeros: 1^*

One Zero: 1^*01^*

Two Zeros: $1^*01^*01^*$

Therefore, $1^* + 1^*01^* + 1^*01^*01^*$

$$c) a + a(a+b)^*a$$

$$Q11) a) (ab+c^*)^*b$$

transition,

$$q_0 \xrightarrow{\epsilon} q_0$$

$$q_0 \xrightarrow{a} q_1$$

$$q_1 \xrightarrow{b} q_0$$

$$q_0 \xrightarrow{b} q_F$$

$$b) a + bb + bab^*a$$

transition,

$$q_0 \xrightarrow{a} q_F$$

$$q_0 \xrightarrow{b} q_1$$

$$q_1 \xrightarrow{b} q_F$$

$$q_0 \xrightarrow{b} q_2$$

$$q_2 \xrightarrow{a} q_3$$

$$q_3 \xrightarrow{b} q_3$$

$$q_3 \xrightarrow{a} q_F$$

$$c) (ab+a)^*(aa+b)$$

Transition,

$$q_0 \xrightarrow{a} q_0$$

$$q_0 \xrightarrow{a} q_1$$

$$q_1 \xrightarrow{b} q_0$$

then final part,

$$q_0 \xrightarrow{a} q_2$$

$$q_2 \xrightarrow{a} q_F$$

and, $q_0 \xrightarrow{b} q_F$

d) $(a*b + b*a)*a$

first branch,

$$p \xrightarrow{a} p$$

$$p \xrightarrow{b} q$$

Second branch,

$$r \xrightarrow{b} r$$

$$r \xrightarrow{a} s$$

Use ϵ transitions,

$$q \xrightarrow{\epsilon} p_0$$

$$s \xrightarrow{\epsilon} p_0$$

and

$$p_0 \xrightarrow{\epsilon} p$$

$$p_0 \xrightarrow{\epsilon} r$$

$$p_0 \xrightarrow{\epsilon} t$$

Finally,

$$t \xrightarrow{a} q_F$$

$$a) \quad R = Q + RP$$

$$E \notin P$$

$$\boxed{R = QP^*}$$

$$\text{Put, } R = QP^*$$

$$\text{then } Q + RP = Q + QP^*P$$

$$P^*P = PP^*$$

$$= Q + QPP^*$$

$$= Q \bullet (E + PP^*)$$

but,

$$E + PP^* = P^*$$

$$Q + RP = QP^* = R$$

$$\boxed{R = QP^*}$$

$$b) (0^*1^*)^* = (0+1)^*$$

$$0 = 0^*1^*$$

$$1 = 0^*1^*$$

$$(0^*1^*)^* = (0+1)^*$$

Hence Proved

$$c) (ab)^* = (a^*b^*)^*$$

take, aabb

then, $aabb \in (a^*b^*)^*$

$$aabb = a^2b^2$$

$$aabb \notin (ab)^*$$

$\therefore \epsilon, ab, abab, ababab, \dots$

$$(ab)^* \neq (a^*b^*)^*$$

d) $(r+s)^* = r^* + s^*$

take, $r=a, s=b$

string ab belongs to $(a+b)^*$ but

$$ab \notin a^* + b^*$$

$$\therefore, (r+s)^* \neq r^* + s^*$$

e) $\lambda + 1^*(011)^*(1^*(011)^*)^*$

is claimed to equal: $(1+011)^*$

$$\text{LHS} \subseteq (1+011)^*$$

every string formed from 1 and 011 ,

$$1^*(011)^*1^*(011)^* \dots$$

$$\therefore \lambda + 1^*(011)^*(1^*(011)^*)^* = (1+011)^*$$

a) From graphy

$$R_1 = aR_1 + bR_2$$

$$R_2 = \epsilon + bR_2$$

$$R_2 = b^*$$

$$R_1 = aR_1 + bb^*$$

$$R_1 = a^* bb^*$$

$$\boxed{RE = a^* bb^*}$$

OR

$$\boxed{a^* b^+}$$

b) $RE = E$

Q14) From the diagram:

q_1 is initial

q_1 has loop 0

$q_1 \xrightarrow{1} q_2$

q_2 is final

q_2 has loop 0

$q_2 \xrightarrow{1} q_3$

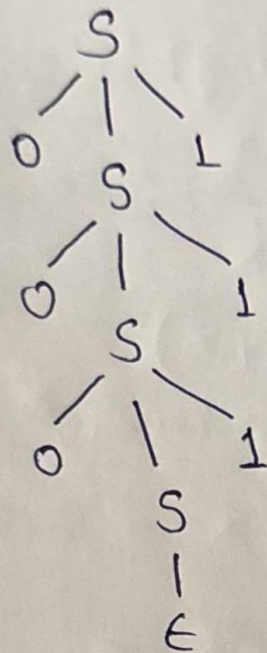
q_3 is dead state with 0, 1 loops

$$\therefore 0^* 1 0^*$$

Q15) i) $L_1 = \{0^n 1^n : n \geq 0\}$

$$S \rightarrow 0S1 \mid \epsilon$$

Parse tree

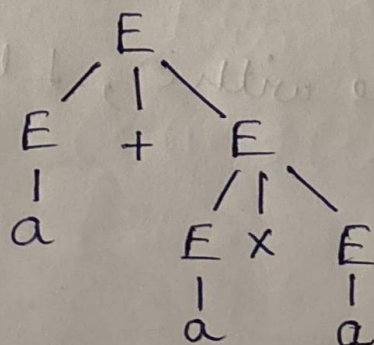


= 000111

ii) a) Use , $S \rightarrow (s)S \mid \epsilon$

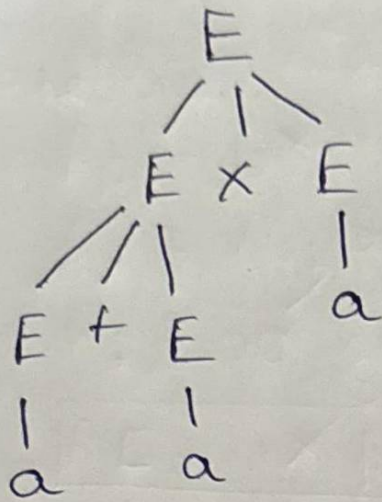
Examples: $()$, $(())$; $()()$ all are generated

b) $E \rightarrow E + E \mid E \times E \mid (E) \mid a$
for $a + a \times a$



$$\Rightarrow a + (a \times a)$$

Parse 2



$$\Rightarrow (a + a) \times a$$

A grammar is unambiguous

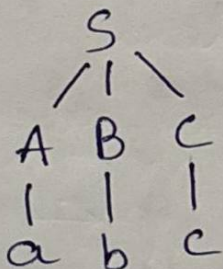
Q16) $S \rightarrow ABC$

$$A \rightarrow a$$

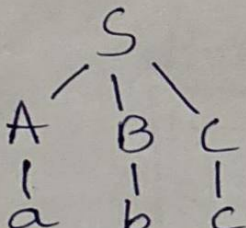
$$B \rightarrow b$$

$$C \rightarrow c$$

LMD



RMD

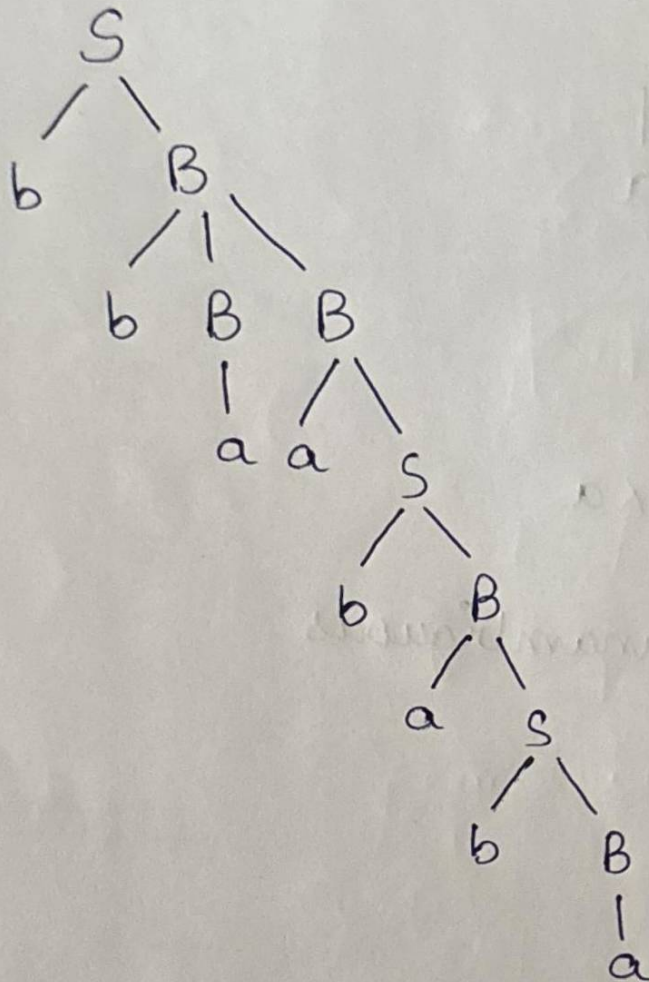


Q17) $S \rightarrow bB|aA$

$A \rightarrow b|bs|aAA$

$B \rightarrow a|as|bBB$

LMD



$\Rightarrow$ bbababab

RMD

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$$b) E \rightarrow E + E \mid E \times E \mid a$$

$$\boxed{\begin{array}{l} E \rightarrow a E' \\ E' = E E' \mid \times E E' \mid \epsilon \end{array}}$$

$$c) E \rightarrow E + T \mid T$$

$$T \rightarrow T \times F \mid F$$

$$F \rightarrow id$$

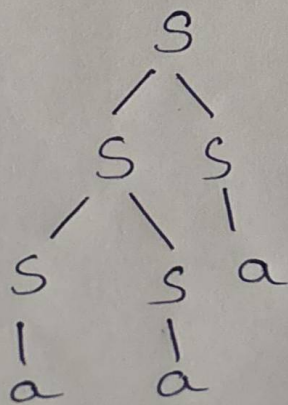
$$\boxed{\begin{array}{l} E \rightarrow T E' \\ E' \rightarrow T E' \mid \epsilon \end{array}}$$

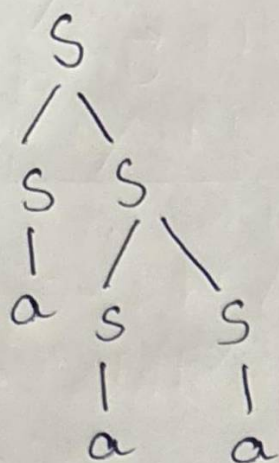
$$\boxed{\begin{array}{l} T \rightarrow F T' \\ T' \rightarrow \times F T' \mid \epsilon \end{array}}$$

$$\boxed{F \rightarrow id}$$

$$Q19) a) S \rightarrow SS \mid a \mid b$$

two different parse tree,





$\therefore$, Ambiguous

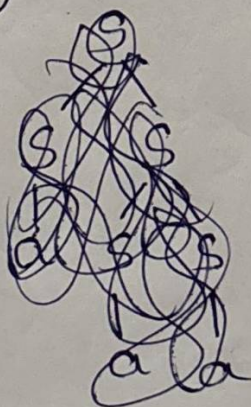
- 6) $S \rightarrow A|B$
 $A \rightarrow aAb|ab$
 $B \rightarrow abB|\epsilon$

first derivation,

S
 $\downarrow$
 A
 $\downarrow$
 ab

$\Rightarrow ab$

Second,



S
 $\downarrow$
 B
 $\downarrow$
 a

$\Rightarrow ab$

Q20)

$$a) S \rightarrow XB|AA$$

$$A \rightarrow a|SA$$

$$B \rightarrow b$$

$$X \rightarrow a$$

$$\text{CNF: } A \rightarrow BC$$

$$A \rightarrow a$$

$\therefore$ the grammar is in CNF

$$b) S \rightarrow a|aA|B$$

$$A \rightarrow aBB|\epsilon$$

$$B \rightarrow Aa|b$$

① Simplification : Remove null production, unit & Useless

$$S \rightarrow a|aA$$

$$A \rightarrow aBB$$

$$B \rightarrow Aa|b$$

② Convert to CNF

$$S \rightarrow a|b|XA|AX$$

$$A \rightarrow XC$$

$$B \rightarrow AX|a|b$$

$$C \rightarrow BB$$

$$X \rightarrow a$$

$$a) L = \{a^n b^{2n} : n \geq 1\}$$

for every push two x's

for every b pop one x

Accept when all x's are removed,

Transitions

$$\delta(q_0, a, z_0) = (q_0, xxz_0)$$

$$\delta(q_0, a, x) = (q_0, xxx)$$

for first b

$$\delta(q_0, b, x) = (q_1, \epsilon)$$

$$\text{then, } \delta(q_1, b, x) = (q_1, \epsilon)$$

$$\delta(q_0, \epsilon, z_0) = (q_f, z_0)$$

$$b) L = \{0^n 1^m 0^n : m, n \geq 1\}$$

Transitions

$$\delta(q_0, 0, z_0) = (q_0, xz_0)$$

$$\delta(q_0, 0, x) = (q_0, xx)$$

Start 1 block,

$$\delta(q_0, 1, x) = (q_1, x)$$

$$\delta(q_1, 1, x) = (q_1, x)$$

Start final 0,

$$\delta(q_1, 0, X) = (q_2, \epsilon)$$

$$\delta(q_2, 0, X) = (q_2, \epsilon)$$

Accept,

$$\delta(q_2, \epsilon, Z_0)$$

$$Q22) L = \{0^n 1^n : n \geq 1\}$$

$$P = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$$

$$\text{where, } Q = \{q_0, q_1, q_2\}$$

$$\Sigma = \{0, 1\}$$

$$\Gamma = \{X, Z_0\}$$

$$F = \{q_F\}$$

Transitions

for every 0, Push X

$$\delta(q_0, 0, Z_0) = (q_0, X Z_0)$$

$$\delta(q_0, 0, X) = (q_0, X X)$$

$$\delta(q_0, 1, X) = (q_1, \epsilon)$$

$$\delta(q_1, 1, X) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, Z_0) = (q_F, Z_0)$$

$$c) \quad n_0 = 2n_1$$

For input 1, If stack has X:

$$X \rightarrow \epsilon$$

If stack is Z_0 or contains Y:

$$Z_0 \rightarrow YZ_0$$

$$Y \rightarrow YY$$

For input 0, If stack has X:

$$X \rightarrow XXX$$

If stack has exactly one Y:

$$Y \rightarrow X$$

Accept:

$$\text{Stack} = Z_0$$

d) Transitions while pushing:

$$\delta(q_0, 0, X) = (q_p, 0X)$$

Push X for 0

$$\delta(q_p, 1, X) = (q_p, 1X)$$

Matching:

$$0/X \rightarrow \epsilon$$

$$1/X \rightarrow \epsilon$$

$$e) L = \{a^m b^{m+n} c^n : m, n \geq 1\}$$

a-block:

$$\delta(q_0, a, z_0) = (q_0, xz_0)$$

$$\delta(q_0, a, x) = (q_0, xx)$$

First m b's:

$$\delta(q_0, b, x) = (q_1, \epsilon)$$

$$\delta(q_1, b, x) = (q_1, \epsilon)$$

Remaining n b's:

$$\delta(q_1, b, z_0) = (q_2, yz_0)$$

$$\delta(q_2, b, y) = (q_2, yy)$$

c block:

$$\delta(q_2, c, y) = (q_3, \epsilon)$$

$$\delta(q_3, c, y) = (q_3, \epsilon)$$

Accept when: z_0

continued
for 000111

initial: $(q_0, 000111, z_0)$

After first 0, $(q_0, 001111, xz_0)$

$\delta(q_0, 001111; xxz_0)$

$\delta(q_0, 111, xxxz_0)$

$\delta(q_0, 11, xxxz_0)$

$\delta(q_0, 1, xz_0)$

$\delta(q_0, \epsilon, z_0)$

$\delta(q_f, \epsilon, z_0)$

$\therefore 000111$ is accepted

CFCs and PDAs are equivalent in expressive power

Q23) Transition idea:

X = crossed off 0

Y = crossed off 1

B = Blank

Transitions:

$$\delta(q_0, 0) = (q_1, X, R)$$

$$\delta(q_0, X) = (q_0, X, R)$$

$$\delta(q_0, Y) = (q_0, Y, R)$$

$$\delta(q_0, L) = (q_{rej}, L, R)$$

$$\delta(q_0, B) = (q_{acc}, B, R)$$

Search state:

$$\delta(q_1, 0) = (q_1, 0, R)$$

$$\delta(q_1, X) = (q_1, X, R)$$

$$\delta(q_1, Y) = (q_1, Y, R)$$

$$\delta(q_1, L) = (q_2, Y, L)$$

$$\delta(q_1, B) = (q_{rej}, B, R)$$

Return State:

$$\delta(q_2, X) = (q_2, X, L)$$

$$\delta(q_2, 0) = (q_2, 0, L)$$

$$\delta(q_2, Y) = (q_2, Y, L)$$

$$\delta(q_2, B) = (q_0, B, R)$$

Input 0011:

Eventually:

XYX

all symbols are matched

Turing decidable

- accepts every string in the language
- rejects every string not in the language
- and halts on every input

eg:-

$$\{0^n 1^n : n \geq 0\}$$

Turing recognizable

- accepts string in the language
- but may loop forever for strings not in the language

eg:-

$$A_{TM}$$

Multiple-tape TMs:

A multi tape TM does not have greater computational power than a single tape TM

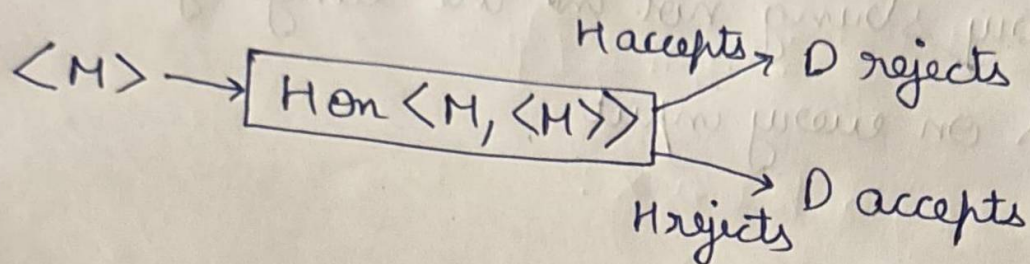
Therefore,

$$\text{Multi tape TM} \equiv \text{Single tape TM}$$

Church Turing Thesis:

Every effectively computable function can be computed by a Turing machine.

Q24) a) machine D



b)

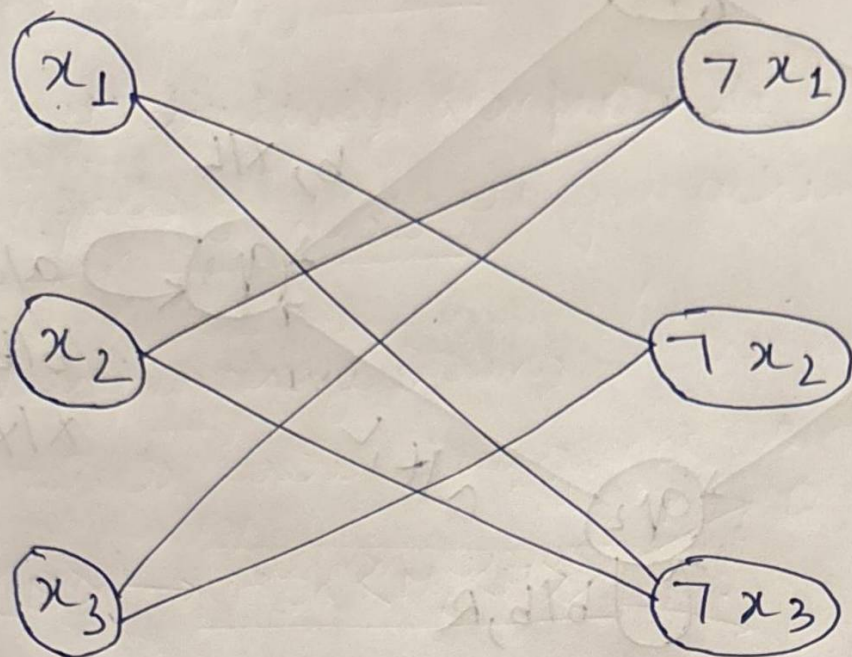
P: consists of problems solvable in polynomial time

NP: proposed solution can be verified in polynomial time

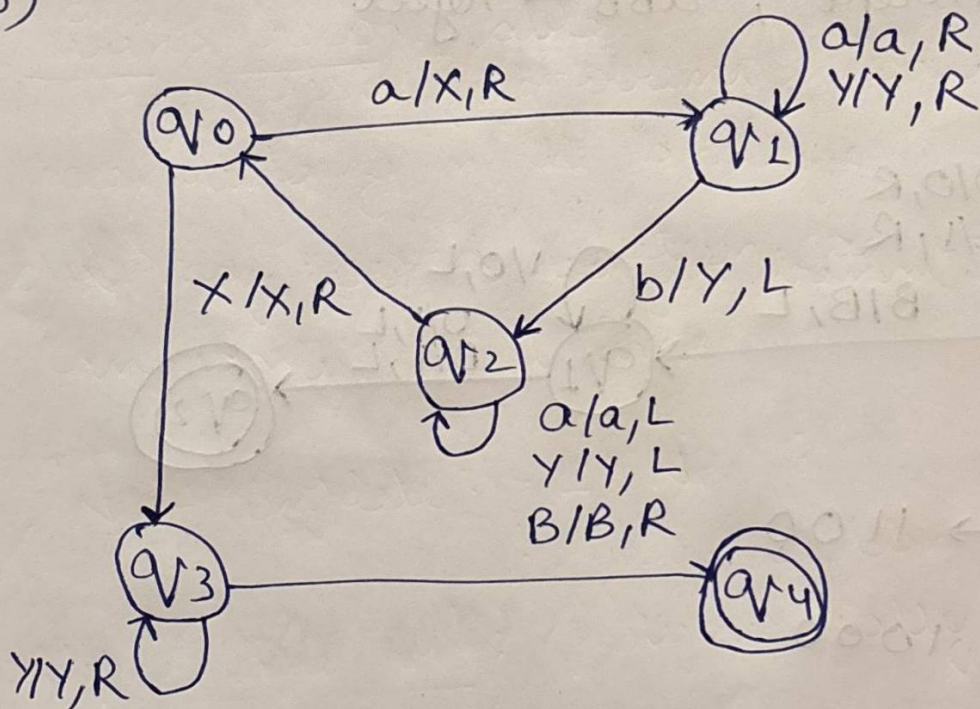
NP-Complete: • L ∈ NP

• Every problem in NP can be polynomially reduced to L

Cook-Levin Theorem: SAT is NP-complete

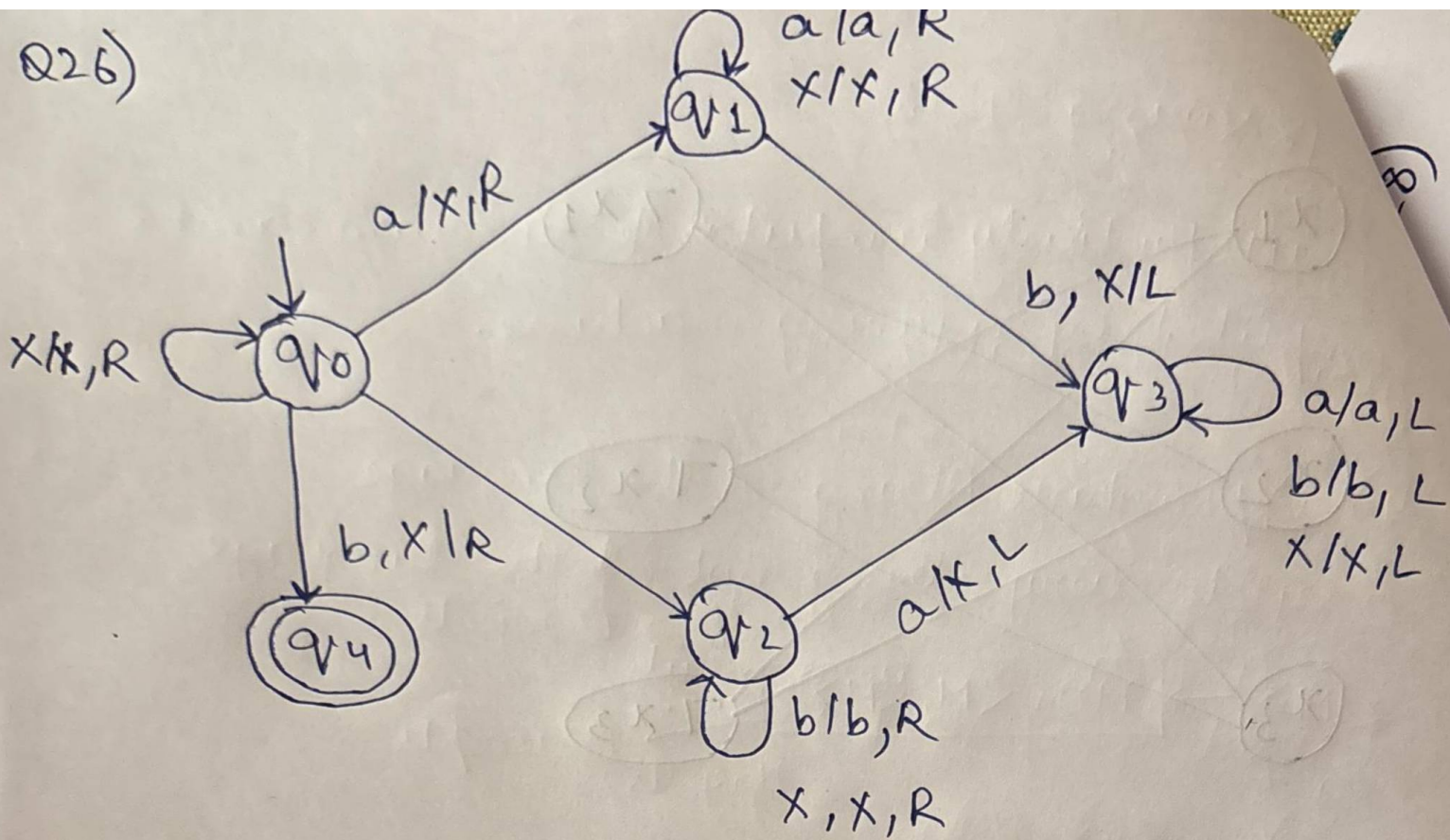


Q25)



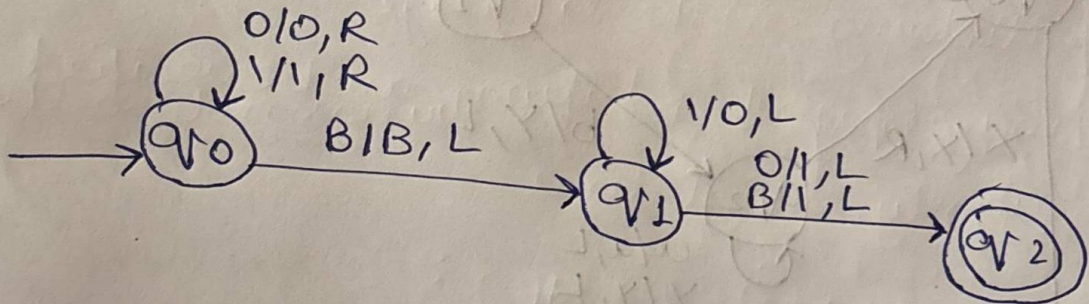
$aaabbb \rightarrow \text{Accept}$
 $aabbbb \rightarrow \text{reject}$
 $aabbab \rightarrow \text{reject}$

Q26)



abba → Accept, abb → reject

Q27)

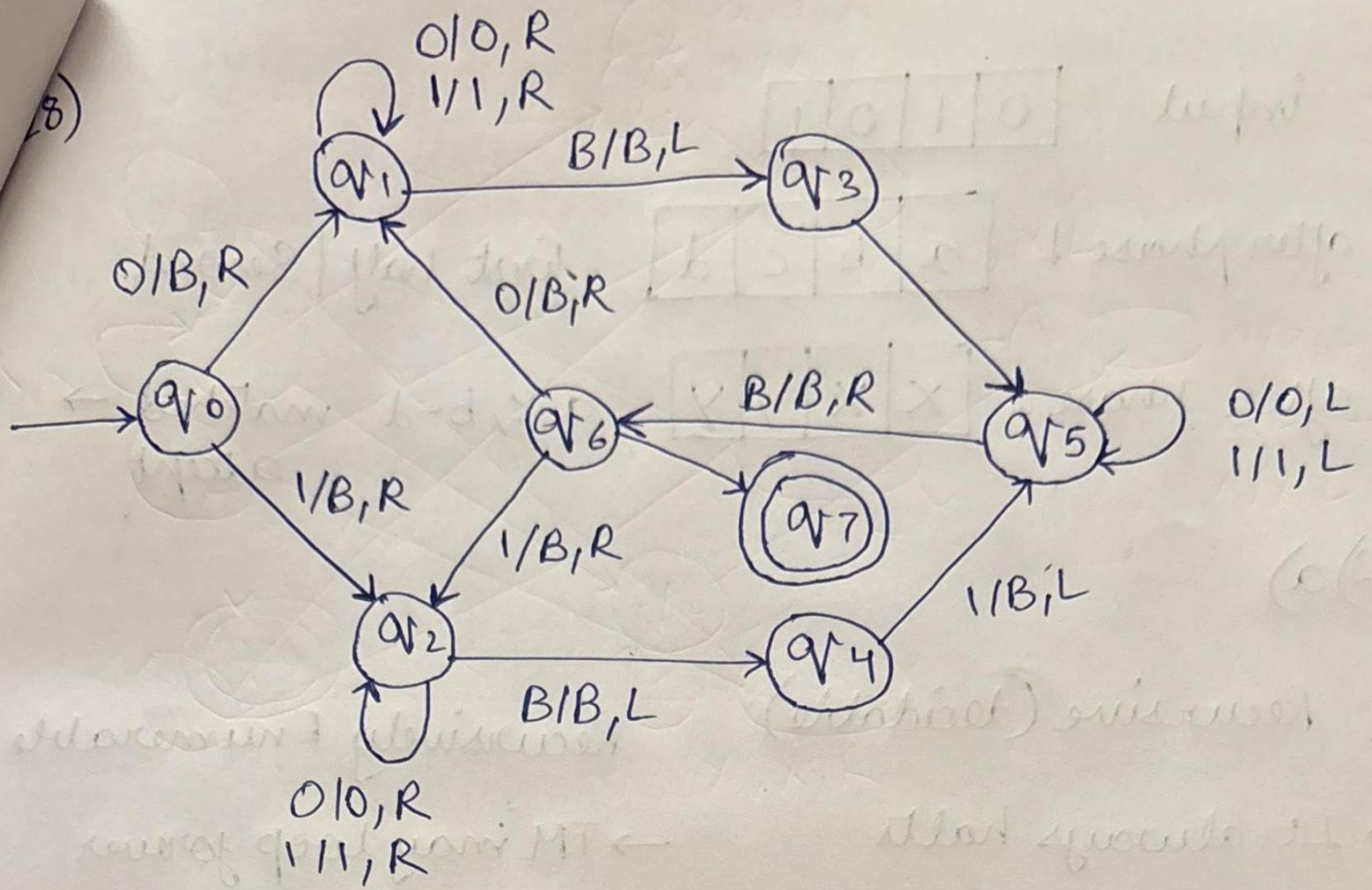


1101 → 1100

111 → 1000

11001 → 11000
 11011 → 11000
 11010 → 11000

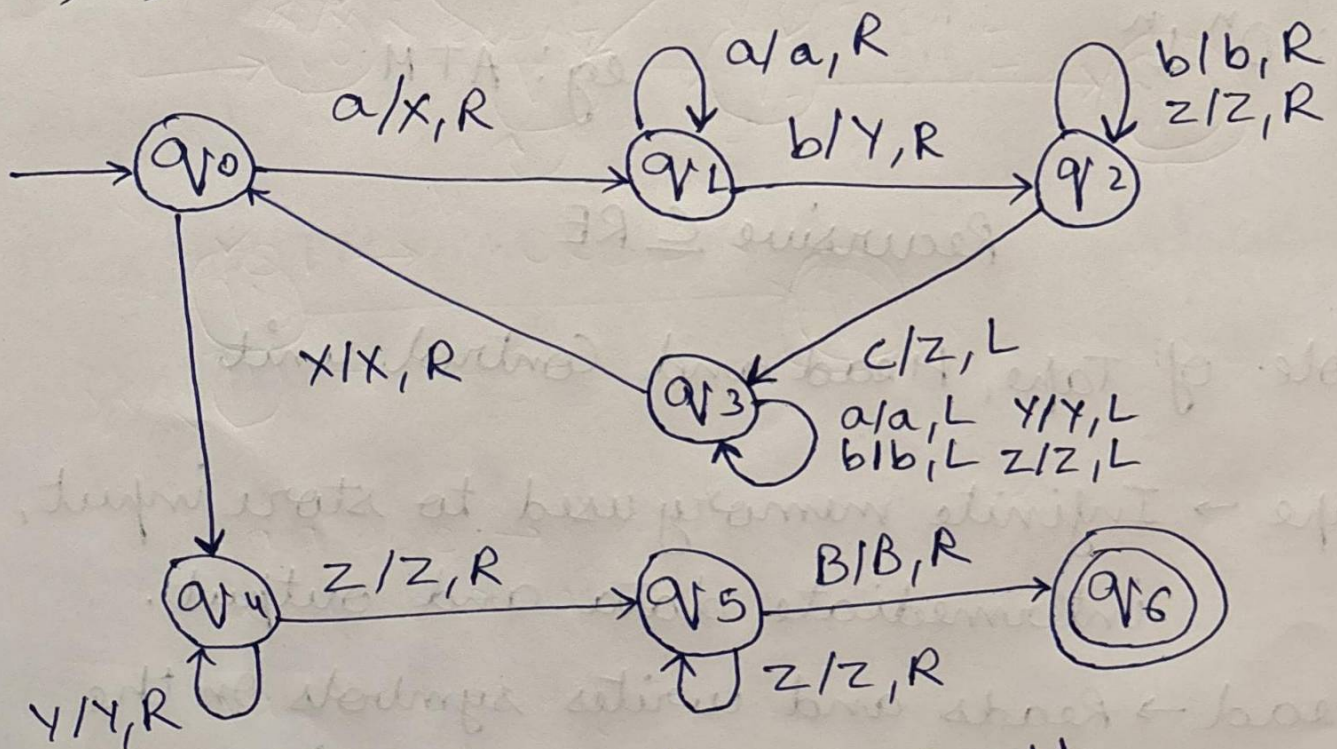
28)



0110 → accept

010 → reject

Q29) a)



aabbcc → accept

b)

input

0	1	0	1
---	---	---	---

after phase 1

a	b	c	d
---	---	---	---

 first half | Second

after phase 2

x	x	y	y
---	---	---	---

 a-c, b-d matched → accept

Q30) a)

Recursive (decidable)

Recursively Enumerable

- | | |
|---------------------------|--|
| → It always halts | → TM may loop forever |
| → Accepts string in L | → Accepts string in L |
| → Rejects string not in L | → May reject or loop for string not in L |

eg: $0^n 1^n$

eg: ATM

Recursive $\subseteq$ RE

b) Role of Tape, Head and Control unit

Tape → Infinite memory used to store input, intermediate data and output.

Head → Reads and writes symbols on the tape and moves left or right

Control Unit → stores the current state and determines the next transition based on the current symbol

c) How a TM simulates any computer algorithm:-

A TM can represent

- Input / data → Tape Symbols
- Memory → Tape
- Instructions → states and transitions
- loops → repeated transitions
- Conditions → different transitions
- output → Tape contents

Thus, a TM can simulate any effectively computable algorithm